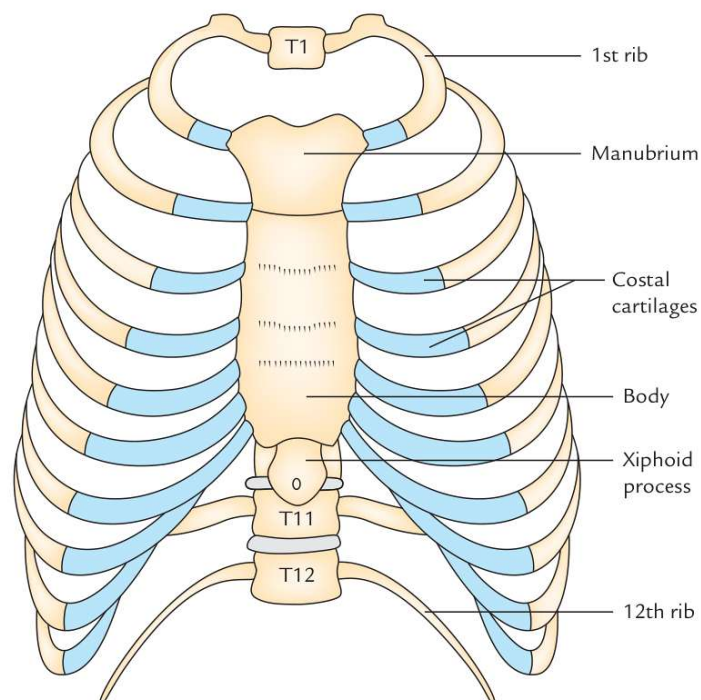


Thorax and Mechanism of Respiration

Core idea

The thorax is the upper part of the trunk. It encloses the thoracic cavity, protects the heart and lungs, and forms a movable osteocartilaginous cage that changes its diameters during breathing.

Main framework	Cavity division	Main respiratory muscle	Movement principle
Sternum, 12 pairs of ribs, costal cartilages and 12 thoracic vertebrae	Two pleural cavities + mediastinum	Diaphragm with help from intercostal muscles	Increase vertical, anteroposterior and transverse diameters



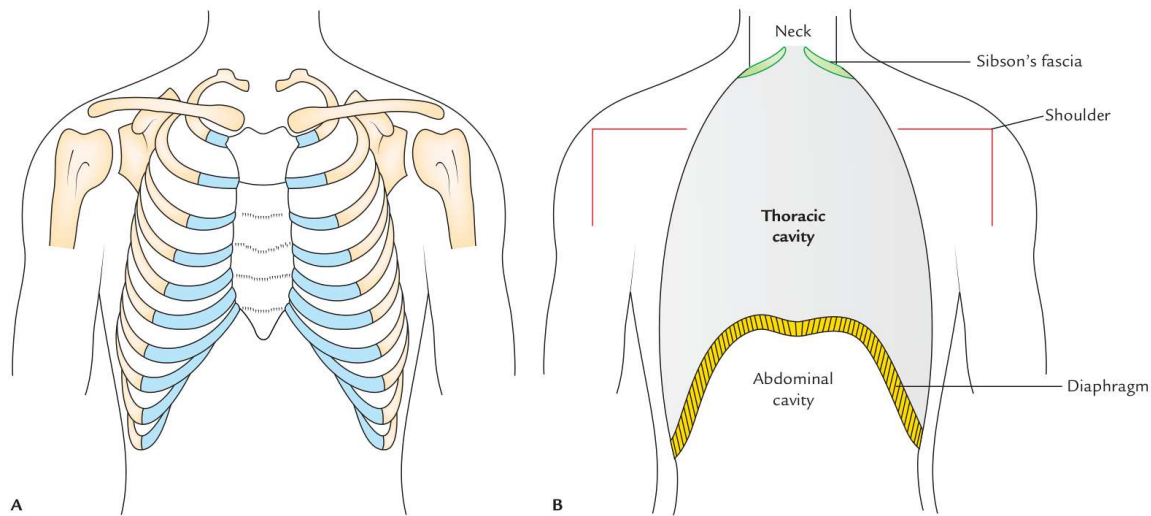
Source diagram: anterior aspect of the thoracic cage.

Image explanation

The thoracic cage is made by the sternum anteriorly, thoracic vertebrae posteriorly, and ribs with costal cartilages laterally. The upper seven ribs reach the sternum directly, ribs 8-10 form the costal margin through cartilage, and ribs 11-12 are floating ribs.

1. THORACIC CAGE - FORMATION AND SHAPE

The thoracic cage is elastic and dynamic rather than a rigid box. Its elasticity comes mainly from the costal cartilages, while its movements occur at costovertebral, costotransverse, sternocostal and interchondral joints.



Source diagram: shape of thoracic cage and thoracic cavity.

Image explanation

The apparent upper breadth of the thorax is increased by the shoulder girdle, while the true thoracic cavity is smaller because the diaphragm projects upwards from below and the shoulder region projects inward above.

Formation of thoracic cage

- Anteriorly: sternum and costal cartilages.
- Posteriorly: 12 thoracic vertebrae and intervertebral discs.
- Laterally: 12 pairs of ribs and the intercostal spaces.

Functions

- Protects heart, lungs and great vessels.
- Provides attachment to muscles of thorax, back, upper limb and diaphragm.
- Changes volume of thoracic cavity during respiration.

2. THORACIC APERTURES

The thoracic cavity communicates with the root of the neck through the superior thoracic aperture, also called the thoracic inlet by anatomists. Inferiorly, the thoracic outlet is closed by the diaphragm but transmits major structures between thorax and abdomen.

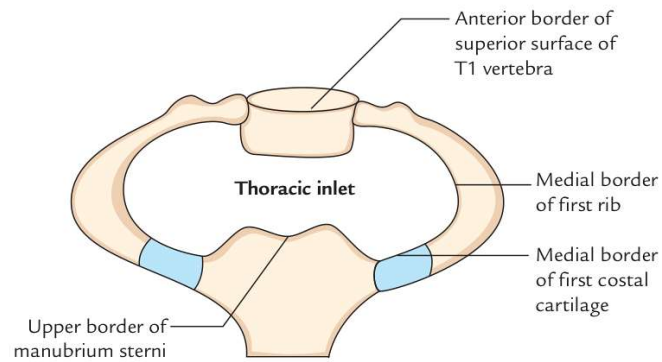
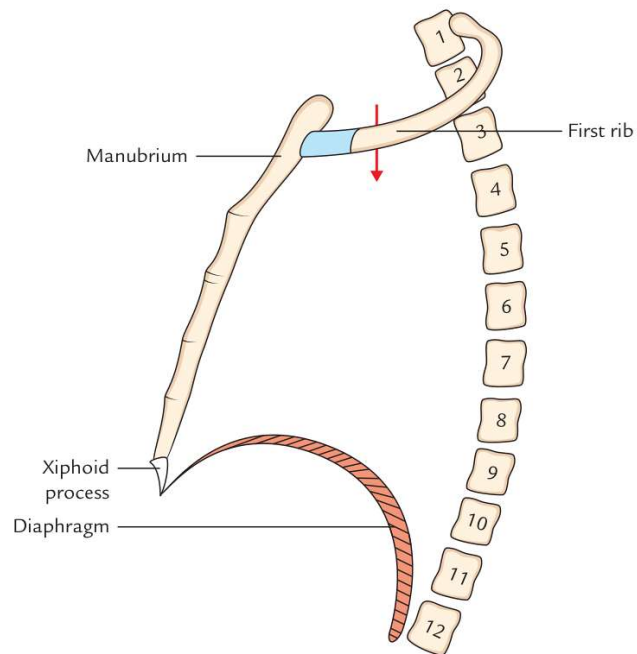


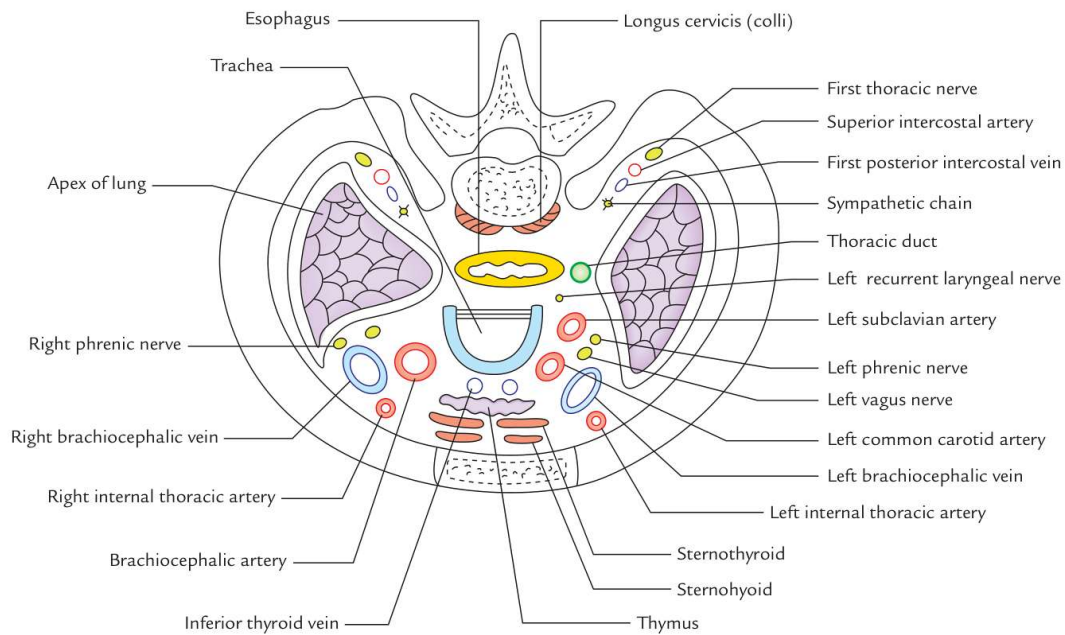
Fig. 14.4 Boundaries of the thoracic inlet.



Source diagrams: boundaries and slope of the thoracic inlet.

Superior thoracic aperture

Boundaries: anteriorly the superior border of manubrium sterni, posteriorly the body of T1 vertebra, and laterally the first rib with its cartilage. The inlet slopes downwards and forwards, so the apex of the lung and cervical pleura project into the root of the neck.

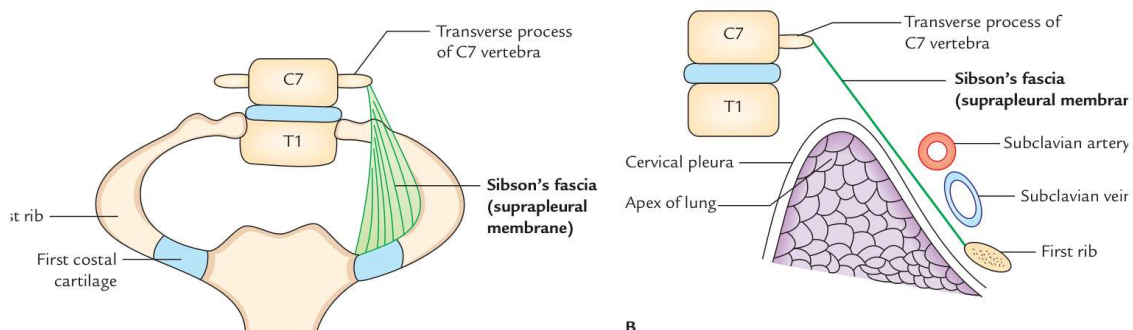


Source diagram: structures passing through the thoracic inlet.

Image explanation

Important structures crossing the inlet include the trachea, esophagus, apices of lungs with cervical pleura, vagus nerves, phrenic nerves, recurrent laryngeal nerve, major arteries, brachiocephalic veins and the thoracic duct. This explains why the thoracic inlet is clinically important in compression syndromes.

3. SUPRAPLEURAL MEMBRANE AND THORACIC INLET SYNDROME



Source diagram: suprapleural membrane or Sibson fascia.

Suprapleural membrane

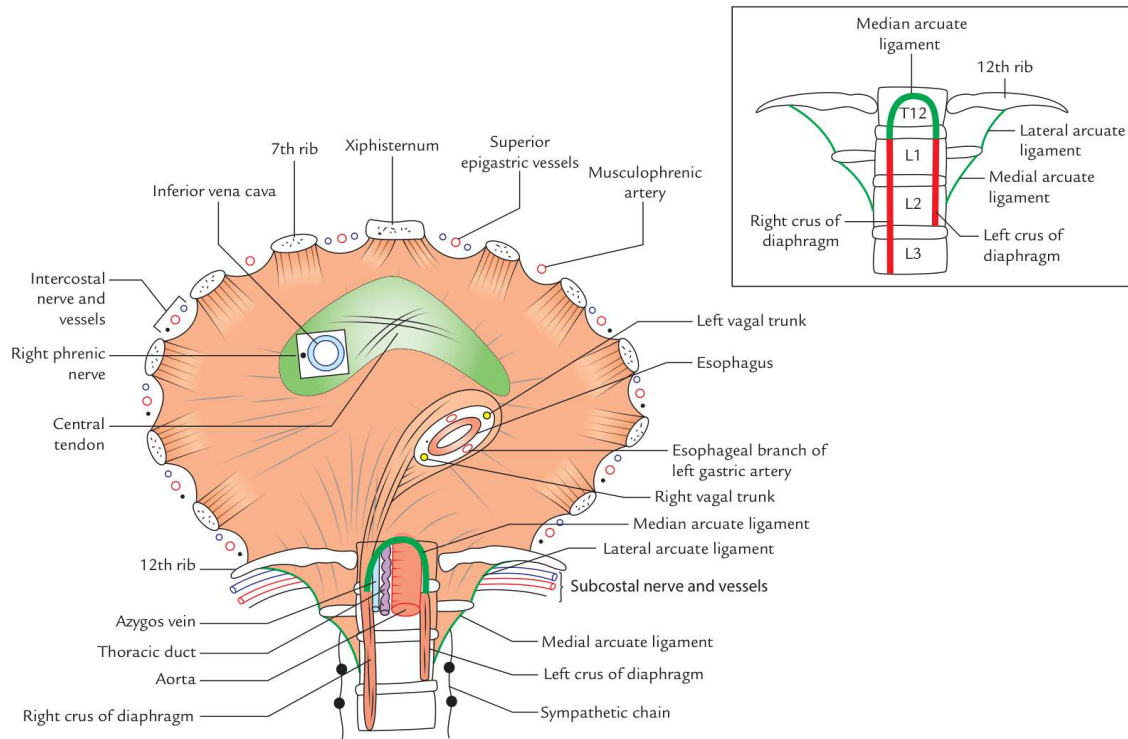
It is a tent-shaped fascial sheet that closes the thoracic inlet on each side. Apex attaches to the transverse process of C7; base attaches to the inner border of first rib and first costal cartilage. It protects the cervical pleura and resists intrathoracic pressure during respiration.

Applied anatomy: thoracic inlet syndrome

Compression of the subclavian artery and lower trunk of brachial plexus over the first rib may cause pain, numbness, tingling and wasting on the medial side of forearm and hand. It may occur in the presence of a cervical rib or tight scalene anterior muscle.

4. DIAPHRAGM OF THE THORACIC OUTLET

The inferior thoracic aperture is closed by the diaphragm. It separates the thoracic and abdominal cavities, but it also provides openings for the inferior vena cava, esophagus, aorta and several smaller structures.



Source diagram: origin, insertion and openings of the diaphragm.

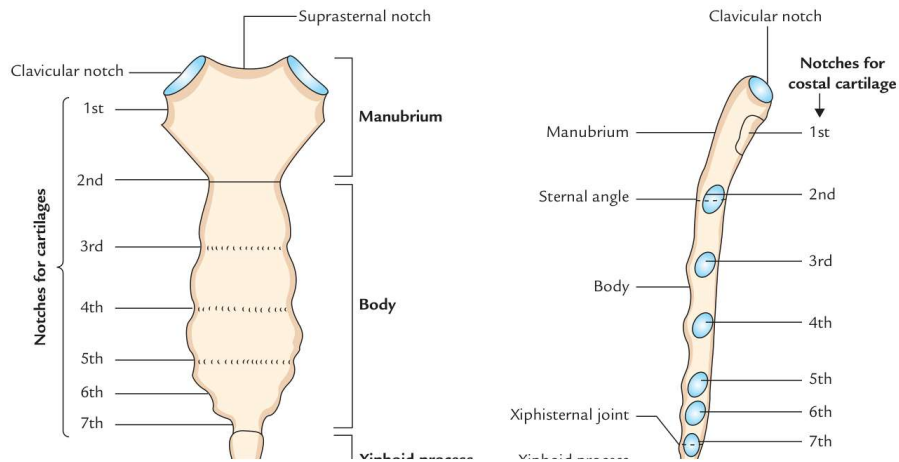
Image explanation

The diaphragm has sternal, costal and vertebral origins. The central tendon is the insertion. The caval opening is in the central tendon at T8, the esophageal opening is in the muscular part at T10, and the aortic opening lies behind the diaphragm at T12.

Opening	Vertebral level	Main structures
Caval opening	T8	Inferior vena cava; right phrenic nerve.
Esophageal opening	T10	Esophagus, vagal trunks, esophageal vessels.
Aortic opening	T12	Aorta, thoracic duct, azygos vein.

5. BONES OF THE THORAX

The bones of the thorax form the skeletal framework for protection and movement. The sternum lies anteriorly, ribs curve around the thoracic wall, and thoracic vertebrae form the posterior support.



Source diagram: features of sternum - anterior and lateral aspects.

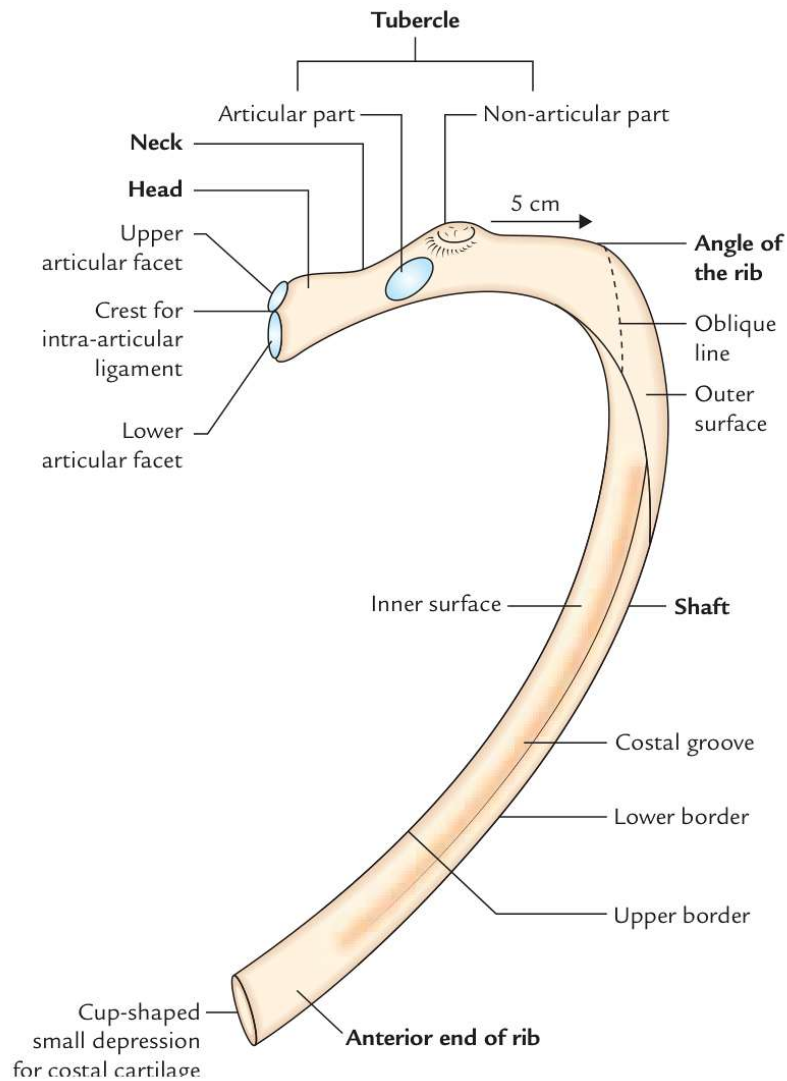
Image explanation

The sternum consists of manubrium, body and xiphoid process. The sternal angle is an important landmark at the level of the second costal cartilage and is used for counting ribs during clinical examination.

Sternum part	High-yield point
Manubrium	Forms the superior part; bears clavicular notches and articulates with the first costal cartilage.
Body	Articulates with costal cartilages and forms most of the anterior thoracic wall.
Xiphoid process	Small inferior part; the xiphisternal joint is a useful surface landmark near the subcostal angle.

Why the sternum is clinically useful

The sternum is central in surface anatomy. The suprasternal notch helps locate the trachea, the sternal angle helps count ribs, and the xiphisternal region helps orient the lower thoracic boundary. These landmarks are repeatedly used during inspection, palpation, percussion and auscultation.

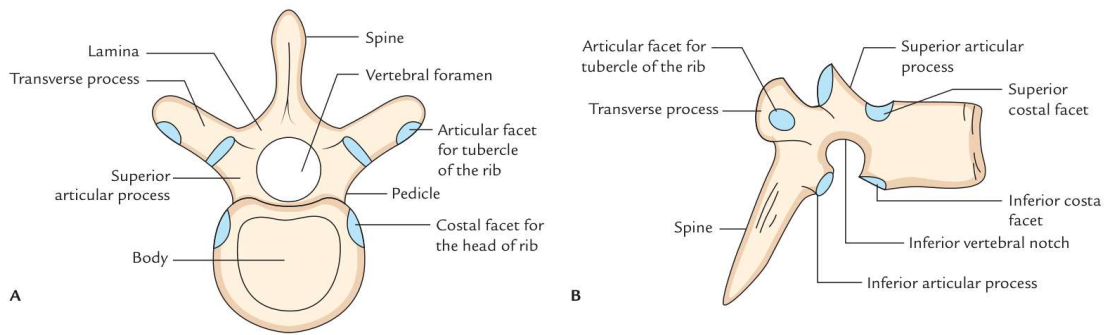


Source diagram: features of a typical rib.

Typical rib

A typical rib has an anterior end, posterior end and shaft. The posterior end has head, neck and tubercle; the shaft has a costal groove near the lower border. The costal groove protects the intercostal vein, artery and nerve, arranged from above downwards as VAN.

6. THORACIC VERTEBRAE AND THORACIC JOINTS



Source diagram: typical thoracic vertebra - superior and lateral views.

Image explanation

Typical thoracic vertebrae have costal facets on the body for the head of rib and facets on transverse processes for the tubercle of rib. Their long, downward sloping spinous processes protect the spinal canal and limit excessive extension.

Important joints

- Costovertebral joint: head of rib with bodies of thoracic vertebrae.
- Costotransverse joint: tubercle of rib with transverse process.
- Sternocostal and interchondral joints: anterior chest wall mobility.

Why these joints matter

- They allow rib elevation and depression.
- Their axes determine pump-handle and bucket-handle movements.
- Respiration depends on coordinated small movements at many rib joints.

Bridge to the rib-vertebra joints

The thoracic vertebra and ribs do not work as isolated bones. Each rib has a posterior lever arm at the vertebral column and an anterior lever arm towards the sternum, so even small movements at the posterior joints can enlarge the thoracic cavity during inspiration.

RIB-VERTEBRA JOINT DIAGRAMS

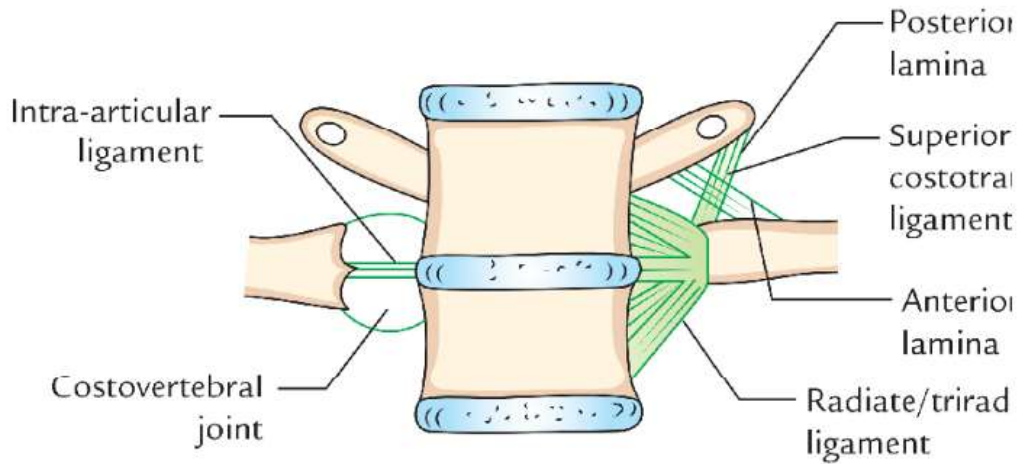
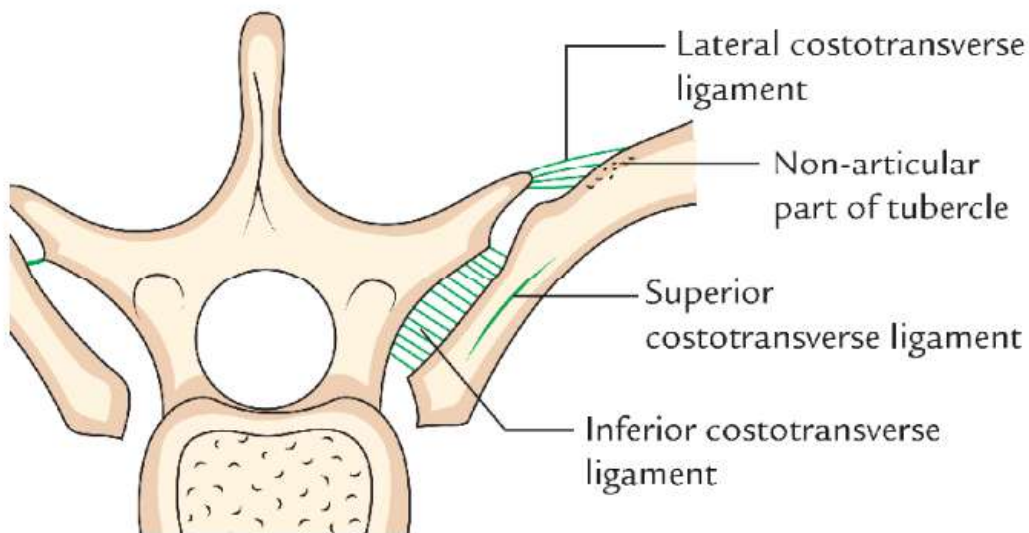


Fig. 15.11 Costovertebral joint.

Source diagram: costovertebral joint.

Costovertebral joint

This joint is formed between the head of a rib and the bodies of adjacent thoracic vertebrae with the intervening intervertebral disc. Its radiate and intra-articular ligaments stabilize the rib head while still allowing the small gliding movements needed during respiration.



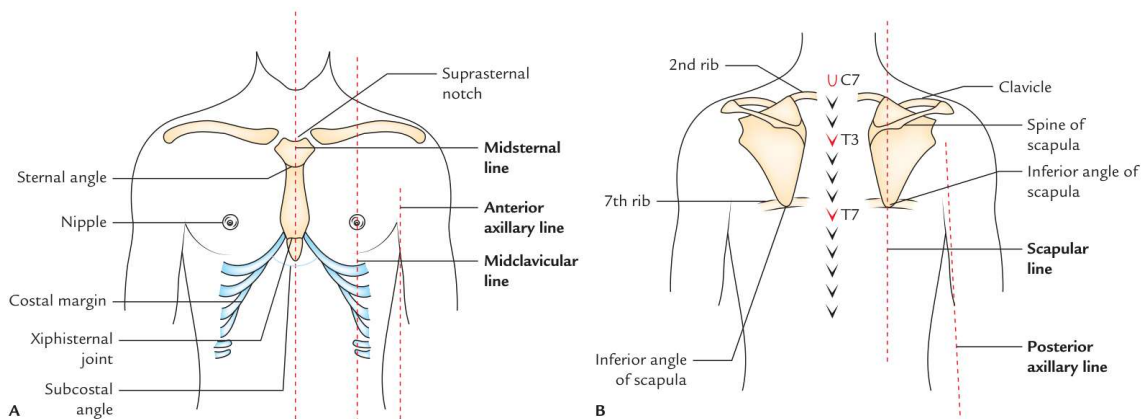
Source diagram: costotransverse joint.

Image explanation

The costovertebral joint permits movement of the rib head on vertebral bodies, while the costotransverse joint guides the tubercle of the rib on the transverse process. Together they form the posterior mechanical axis for elevation and depression of the ribs.

Joint	Main ligament support	Respiratory value
Costovertebral	Capsular, radiate and intra-articular ligaments.	Allows small gliding/rotation of the rib head.
Costotransverse	Superior, inferior and lateral costotransverse ligaments.	Stabilizes the rib tubercle and guides rib elevation.

7. SURFACE LANDMARKS AND LINES OF ORIENTATION



Source diagram: surface landmarks on the thoracic wall - anterior and posterior aspects.

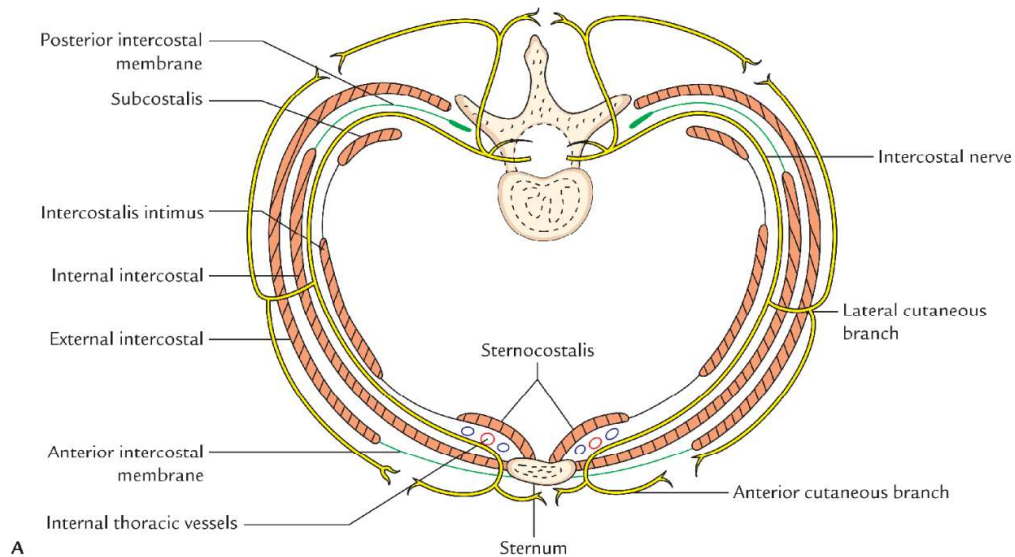
Image explanation

The suprasternal notch, sternal angle, costal margin, subcostal angle, nipple, scapular spine and inferior angle of scapula are practical landmarks. Imaginary vertical lines such as midsternal, midclavicular, anterior axillary, midaxillary, posterior axillary and scapular lines help describe positions of viscera and auscultation points.

Landmark	Clinical use
Suprasternal notch	Trachea can be palpated here; lies near lower border of T2.
Sternal angle	Marks level of second rib; used to count ribs and intercostal spaces.
Xiphisternal joint	Level of upper border of T9; related to subcostal angle.
Inferior angle of scapula	Usually corresponds to seventh rib or seventh intercostal space.

8. THORACIC WALL AND INTERCOSTAL MUSCLES

The thoracic wall is formed by vertebral column posteriorly, sternum and costal cartilages anteriorly, and ribs with intercostal spaces laterally. Its floor is diaphragm and roof is suprapleural membrane.



Source diagram: intrinsic muscles of the thoracic wall and course of typical intercostal nerve.

Image explanation

The intrinsic muscle layers are arranged from superficial to deep as external intercostal, internal intercostal and innermost intercostal. The neurovascular bundle lies between internal intercostal and innermost intercostal muscles.

Intercostal muscle layers

- External intercostal: fibres run downwards, forwards and medially; active in inspiration.
- Internal intercostal: fibres run upwards, forwards and medially; mainly active in expiration.
- Innermost intercostal: deepest layer; separated from internal intercostal by neurovascular bundle.

Functional role

- Support intercostal spaces and prevent bulging.
- Elevate ribs during inspiration.
- Stabilize ribs during forced respiratory movements.

Layer logic

The three intercostal muscle layers are important because the neurovascular bundle lies in the neurovascular plane between internal intercostal and innermost intercostal muscles. This is also why procedures through the intercostal space must respect the costal groove and the upper border of the lower rib.

Layer	Key point
External intercostal	Best remembered as inspiratory in action.
Internal intercostal	Important in forced expiration.
Innermost intercostal	Deepest layer; neurovascular bundle lies superficial to it.

Procedure memory

In the intercostal space, the main VAN bundle is protected in the costal groove near the inferior border of the rib above. Therefore, during pleural aspiration or chest tube placement, instruments are directed close to the upper border of the lower rib to reduce the chance of injuring the main bundle.

Bundle/branch	Position to remember
Main VAN bundle	Costal groove of rib above.
Collateral branch	Runs along upper border of rib below.

9. INTERCOSTAL SPACE AND NEUROVASCULAR BUNDLE

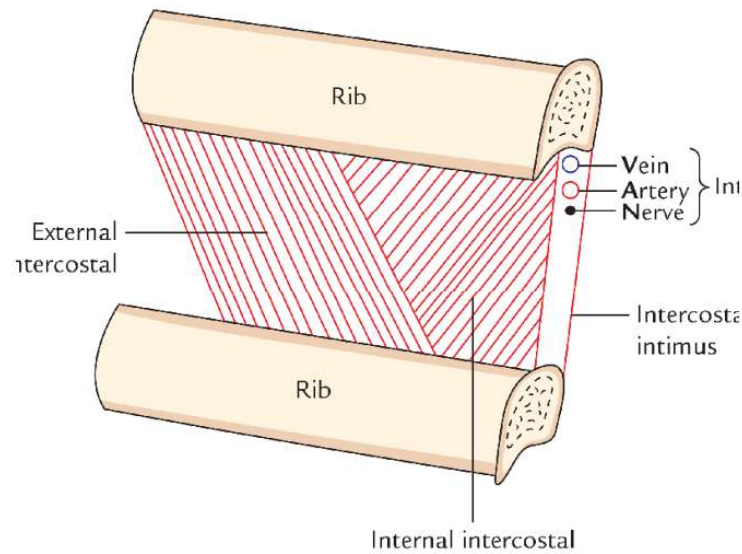
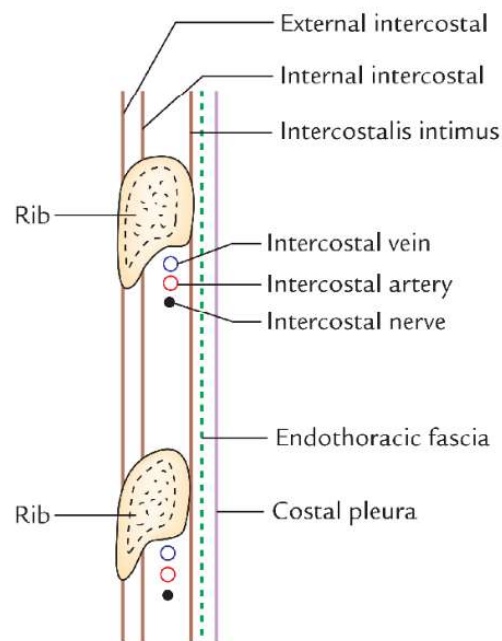


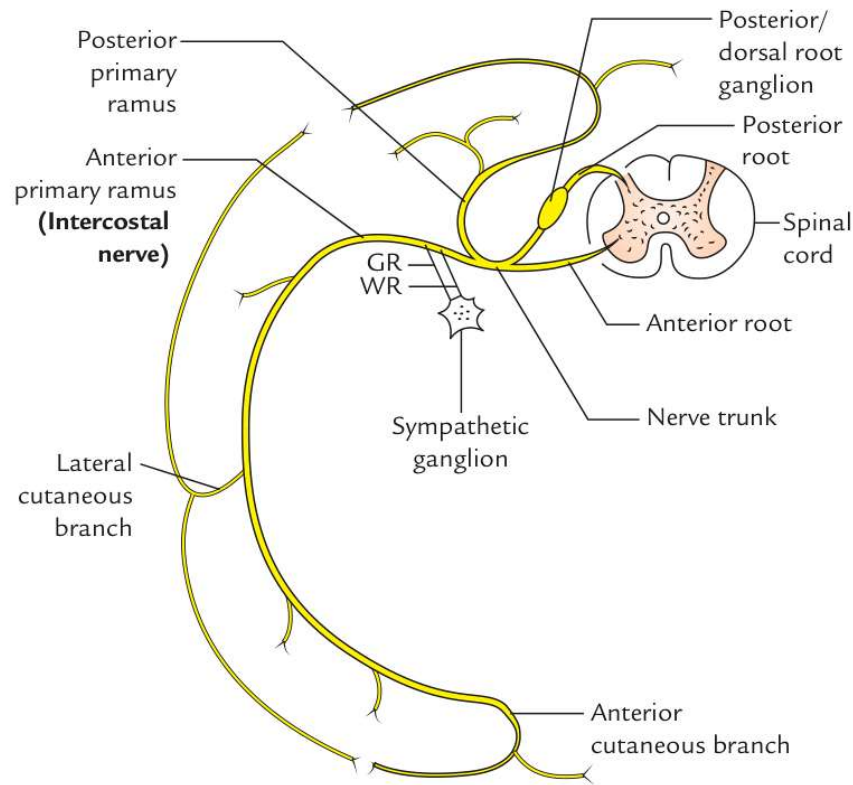
Fig. 16.3 Contents of the intercostal space as seen in dissection.



Source diagrams: contents of intercostal space in dissection and vertical section.

VAN arrangement and contents

The main neurovascular bundle lies in the costal groove between internal intercostal and innermost intercostal muscles. From above downwards it is vein, artery and nerve. A typical intercostal space contains the three intercostal muscle layers, intercostal vessels, intercostal nerve, lymphatics and collateral branches near the upper border of the rib below.



Source diagram: typical thoracic spinal nerve.

Intercostal nerves

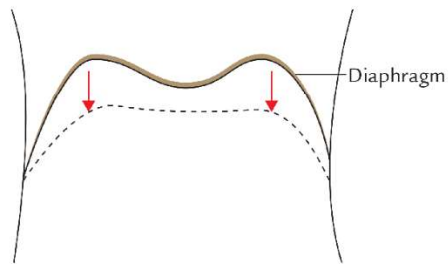
Anterior primary rami of T1-T11 form intercostal nerves, while T12 is the subcostal nerve. They are segmental nerves and usually do not form plexuses. Important branches include muscular branches, collateral branch, lateral cutaneous branch and anterior cutaneous branch.

Applied anatomy: paracentesis thoracis

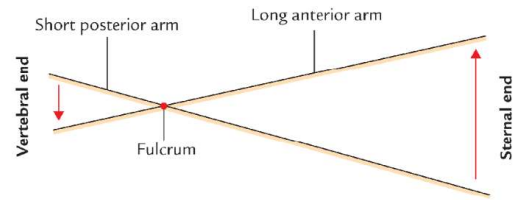
During aspiration of pleural fluid, the needle should not be inserted medial to the angle of the rib because the posterior intercostal artery crosses the space obliquely there. The safer principle is to keep close to the upper border of the rib below.

10. MECHANISM OF RESPIRATION

Respiration consists of inspiration and expiration. During inspiration the thoracic cavity increases in volume and intrathoracic pressure becomes negative, so air is drawn into the lungs. Increase in thoracic capacity occurs in three diameters: vertical, anteroposterior and transverse.



Source diagram: vertical diameter increases by diaphragm descent.



g. 16.11 Increase in anteroposterior diameter of thoracic

Source diagram: lever principle behind AP diameter increase.

Image explanation

Vertical increase is produced mainly by descent of the diaphragm. Anteroposterior increase is produced because each rib acts like a lever: the short posterior arm moves near the vertebral end, while the long anterior arm lifts the sternum forwards and upwards.

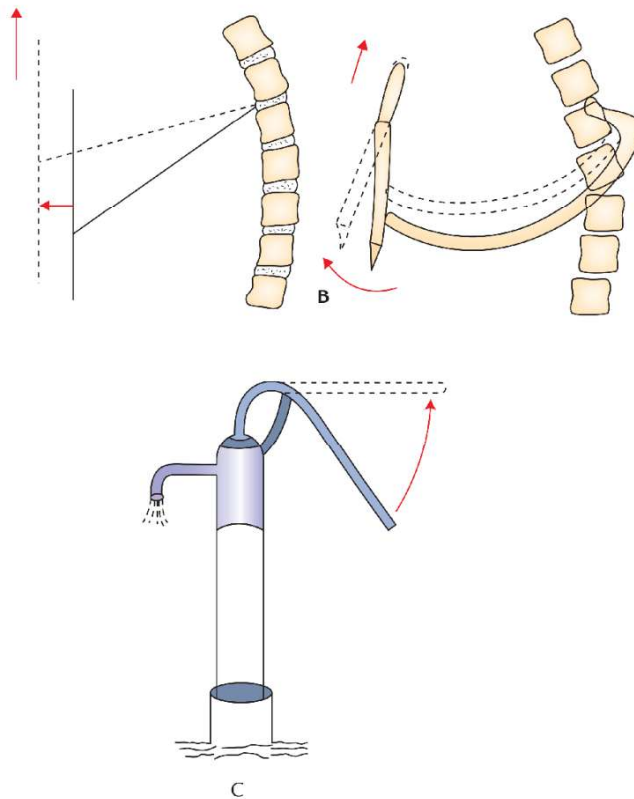
Step 1	Step 2	Step 3	Result
Diaphragm contracts and descends.	Ribs elevate at posterior joints.	Sternum moves forwards; rib shafts move laterally.	Thoracic volume increases and air enters.

Diameter changes in one view

During inspiration, the thoracic cavity enlarges in three directions at the same time. The diaphragm increases the vertical diameter, the sternum moves forwards to increase the anteroposterior diameter, and the ribs move laterally to increase the transverse diameter. These three changes together reduce intrathoracic pressure and draw air into the lungs.

Diameter	Main anatomical movement	Easy memory point
Vertical	Diaphragm contracts and descends.	Floor of thorax moves downwards.
Anteroposterior	Upper ribs and sternum move forwards and upwards.	Pump-handle movement.
Transverse	Middle ribs move upwards and laterally.	Bucket-handle movement.

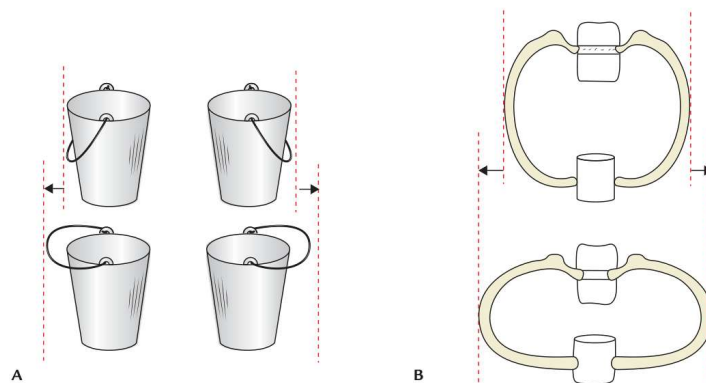
11. PUMP-HANDLE AND BUCKET-HANDLE MOVEMENTS



Source diagram: pump-handle movement of the sternum.

Image explanation

Pump-handle movement mainly increases the anteroposterior diameter. When the upper ribs are elevated, the anterior ends of the ribs and sternum move forwards and upwards like the handle of a water pump.



Source diagram: bucket-handle movement of vertebrochondral ribs.

Image explanation

Bucket-handle movement mainly increases the transverse diameter. During elevation, the shafts of the middle ribs move outwards and upwards like the handle of a bucket, widening the thoracic cavity from side to side.

12. MUSCLES INVOLVED IN RESPIRATION

Type	Inspiration	Expiration
Quiet respiration	Diaphragm and external intercostal muscles.	Passive; no major muscle contraction.
Deep respiration	Diaphragm, external intercostals, scalene muscles, sternocleidomastoid, levatores costarum and serratus posterior superior.	Mostly passive in ordinary deep breathing.
Forced respiration	All muscles of deep inspiration plus accessory muscles fixing and elevating ribs: pectorals, serratus anterior, trapezius, rhomboids and levator scapulae.	Internal intercostals, transversus thoracis, serratus posterior inferior and abdominal wall muscles.

Inspiration summary

- Thoracic volume increases.
- Air enters lungs due to negative intrathoracic pressure.
- Diaphragm descends; ribs elevate; sternum moves forwards and upwards.

Expiration summary

- Quiet expiration is mainly passive.
- Elastic recoil of lungs and relaxation of respiratory muscles reduce thoracic volume.
- Forced expiration uses abdominal and internal intercostal muscles.

13. APPLIED ANATOMY AND EXAM POINTS

Applied point	Why it matters
Asthmatic posture	Patient sits leaning forward with arms fixed. This fixes scapulae so pectoral muscles and serratus anterior can help elevate ribs.
Intercostal nerve block	Local anesthetic is injected near the intercostal nerve to relieve pain from rib fracture or thoracic surgery.
Rib fracture	Sharp rib ends may injure pleura and lung, causing pneumothorax or hemothorax.
Chest tube / pleural aspiration	Inserted carefully with respect to the neurovascular bundle; the main bundle lies in costal groove.
Hiatal hernia	Part of stomach may herniate through esophageal opening of diaphragm.
Coarctation of aorta	Enlarged posterior intercostal arteries may produce rib notching.

Golden facts to remember

Sternal angle helps count ribs. The diaphragm is the chief muscle of quiet inspiration. Main intercostal neurovascular bundle is arranged VAN in the costal groove. Pump-handle movement increases anteroposterior diameter; bucket-handle movement increases transverse diameter.

How to write this answer in exams

Begin with the framework of the thoracic cage, then mention the diaphragm and intercostal muscles, then explain inspiration by the three diameter increases: vertical by diaphragm descent, anteroposterior by pump-handle movement, and transverse by bucket-handle movement. End with expiration and one clinical correlation such as asthma posture or paracentesis thoracis.

Revision trigger	Answer in one line
Most useful landmark	Sternal angle - identifies 2nd rib and helps count intercostal spaces.
Safest procedure principle	Pass instruments close to upper border of lower rib to avoid the main VAN bundle in the costal groove.

Quick viva prompt	Expected answer
Chief muscle of inspiration	Diaphragm.
Order in costal groove	Vein, artery, nerve from above downwards.
Why asthmatic patient leans forward	It fixes the shoulder girdle, allowing accessory muscles to elevate the ribs more effectively.
Connection summary	Rib joints permit elevation, intercostals stabilize the spaces, and the diaphragm changes the vertical dimension.